PROJECT EXPLANATION REPORT

Historical Website Footprinting: Stayzilla

Historical Website Data Collection and Passive OSINT Using the Internet Archive

Project domain: stayzilla.com

Primary source: Internet Archive / Wayback Machine

1. Introduction

Historical Website Footprinting is a passive research activity that uses archived versions of a website to understand what the site looked like and what information it publicly displayed in the past. Websites change over time, and some older pages may no longer be available on the live website. The Internet Archive's Wayback Machine stores snapshots of many websites, allowing researchers to inspect selected historical records.

In this project, Stayzilla (stayzilla.com) is the target domain. The work focuses on locating archived records, recording their URLs and dates, taking screenshots, and describing only what can be observed in those records.

2. Aim of the Project

The aim is to demonstrate how the Wayback Machine can be used for historical website research and passive open-source intelligence (OSINT), while keeping an organized evidence log and avoiding unsupported security claims.

3. Objectives

- Find available historical records for the Stayzilla website.
- Record the archived URL and capture date for each selected record.
- Save screenshots with consistent evidence filenames.
- Describe visible page elements and compare records from different years.
- Explain how historical website information can support passive reconnaissance.
- Document limitations and avoid treating missing archived resources as proof of a vulnerability.

4. Tools and Sources Used

- Internet Archive: a digital library that preserves selected web pages and other digital material.
- Wayback Machine: the Internet Archive service used to search for and open historical website captures.
- Web browser: used to access the archive and inspect the archived pages.
- Screenshots: used to preserve visible evidence for later review.
- Evidence inventory: a table that connects each evidence ID to its URL, date, screenshot filename, and observation.

5. Methodology — How the Project Was Performed

1. **Define the target:** The project selected Stayzilla and recorded stayzilla.com as the domain under review.
2. **Search the archive:** The Wayback Machine calendar was opened to identify dates for which archived records were available.
3. **Open specific snapshots:** Selected archived URLs were opened and their timestamps were used to identify the capture dates.
4. **Capture screenshots:** Screenshots were saved to document the visible page or calendar view. Each screenshot should show enough context to support the observation.
5. **Record evidence:** Each item was assigned an evidence ID and a filename, and the URL, date, and factual observation were added to the Evidence Inventory.
6. **Analyze cautiously:** The visible features in the selected records were described. Any missing images or page resources were treated as archive limitations unless there was separate evidence for another explanation.

6. Explanation of the Evidence

E-01 — Wayback Machine Calendar Overview

The E-01 URL is a wildcard-style Wayback Machine calendar/query URL. It is used to view available historical captures for Stayzilla, rather than opening one specific archived page. For that reason, the capture date is recorded as “N/A — calendar overview.” The calendar helps the researcher choose dates for individual snapshots.

Screenshot filename: E-01_Wayback_Calendar_Overview.png

E-02 — Archived Stayzilla Homepage (26 October 2011)

The E-02 URL contains the timestamp 20111026235702, which corresponds to 26 October 2011. The archived homepage shows hotel-search fields, navigation links such as “ABOUT US” and “CONTACT US,” travel-deal sections, and hotel-booking information. These visible items document elements of the public-facing homepage at the time of this archived capture.

Screenshot filename: E-02_Stayzilla_Homepage.png

E-03 — Archived Stayzilla Website (28 September 2017)

The E-03 URL contains the timestamp 20170928233633, corresponding to 28 September 2017. The selected archived page displays a homestay search interface with destination, check-in, check-out, and traveller fields. Some tourism-partner images appear not to have loaded in the archived version. This may be due to incomplete archived resources and does not, by itself, demonstrate a security vulnerability.

Screenshot filename: E-03_Stayzilla_Homestay_Search.png

7. Findings and Overall Analysis

The selected records show that the Wayback Machine provides a way to examine Stayzilla's publicly visible web content at different points in time. E-01 is a calendar overview for locating captures; E-02 documents homepage features visible in October 2011; and E-03 documents a search interface visible in September 2017. The records can support a historical comparison of website content and layout, but they do not establish why the website changed or whether any current security issue exists.

8. Cybersecurity Relevance

Historical website records can support passive OSINT by helping analysts understand an organization's former public-facing content and digital footprint. In an authorized security assessment, such context may help shape questions for further investigation. Archived content should be treated as historical context, not as proof of a current weakness. This project does not include exploitation, unauthorized access, credential collection, or intrusive scanning.

9. Limitations and Ethical Considerations

- The Wayback Machine may not have saved every page, image, script, or other resource.
- Some archived links may fail, redirect, or display incompletely.
- A capture timestamp indicates when the archive recorded a snapshot; it does not necessarily show when the content was first published.
- Missing images are not, on their own, evidence of a security vulnerability.
- Do not claim that a domain is currently offline or vulnerable without separate, current verification.
- Use archived material responsibly and avoid republishing sensitive personal information.

10. Conclusion

This project demonstrates a basic workflow for historical website footprinting using the Internet Archive's Wayback Machine. The evidence log organizes one calendar overview and two archived Stayzilla page records from 2011 and 2017. The project highlights the value of preserving URLs, dates, screenshots, and factual observations. Its conclusions are limited to the visible archived material; further security conclusions would require separate evidence and appropriate authorization.

11. Short Explanation for a Presentation or Viva

“My project is Historical Website Footprinting using the Internet Archive's Wayback Machine. I selected Stayzilla as the target domain and reviewed historical archived records. I documented three evidence items: a Wayback calendar overview, a Stayzilla homepage snapshot from 26 October 2011, and a Stayzilla page snapshot from 28 September 2017. I recorded each URL, date, screenshot filename, and observation in an evidence inventory. The project is passive OSINT: it examines publicly available archived content and does not attempt to exploit the website. A key limitation is that archived pages may be incomplete, so missing images do not prove a vulnerability.”

12. Evidence Checklist Before Submission

- E-01 screenshot saved as E-01_Wayback_Calendar_Overview.png.
- E-02 screenshot saved as E-02_Stayzilla_Homepage.png.
- E-03 screenshot saved as E-03_Stayzilla_Homestay_Search.png.
- The evidence filenames in the Word report match the actual screenshot filenames.
- Each observation matches what is visible in the corresponding screenshot.
- The original project report and this explanation report are saved in the project folder.



- You can see there is a colour on specific dates which means on that day they post a snapshot
- You have to place your cursor on particular date, after placing it a pop-up will come as show in above screenshot after that click on time, It will show you what they post.

References

Internet Archive: https://archive.org/

Wayback Machine: https://web.archive.org/

Stayzilla archive calendar: https://web.archive.org/web/*/stayzilla.com